

PRBench: A Standardized Probabilistic Robustness Benchmark

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Abstract

Deep learning models are notoriously vulnerable to imperceptible perturbations. Most existing research focuses on adversarial robustness (AR), which evaluates robustness by determining whether a worst-case adversarial example (AE) exists. In contrast, probabilistic robustness (PR) measures the probability that predictions remain correct under stochastic perturbations. While PR is widely regarded as a practical complement to AR, dedicated training methods for improving PR are still relatively underexplored, albeit with emerging progress. Among the few PR-targeted training methods, we identify three limitations: *i*) non-comparable evaluation protocols; *ii*) limited comparisons to adversarial training (AT) baselines despite anecdotal PR gains from AT, and; *iii*) no unified framework to compare the generalization of these methods. Thus, we introduce PRBench, the first benchmark dedicated to evaluating PR performance achieved by different robustness training methods. PRBench empirically compares most common AT and PR-targeted training methods using a comprehensive set of metrics, including clean accuracy, PR and AR performance, training efficiency, and generalization error (GE). We also provide theoretical analysis of the GE across different training methods, grounded in Uniform Algorithmic Stability. Our results reveal two distinct trade-off frontiers: AT methods improve both AR and PR performance at the cost of clean accuracy and GE, whereas PR-targeted methods prioritize high clean accuracy and PR with lower GE while trading off AR performance. Based on these observations, PRBench inspires future

research: subsequent work may benefit from developing versatile, AT-based approaches that achieve balanced performance by jointly enhancing AR and PR while maintaining clean accuracy and low GE. These findings underscore the necessity of PRBench as the first standardized benchmark for PR, complementing the widely studied area of AR. A leaderboard comprising 229 trained models across 7 datasets and 10 model architectures is publicly available at <https://wellzline.github.io/PRBenchLeaderboard/>.

CCS Concepts

• General and reference → Evaluation; Metrics; • Security and privacy → Trust frameworks; • Computing methodologies → Computer vision problems.

Keywords

Probabilistic Robustness, Uniform Algorithmic Stability Theory, Robust Overfitting, Adversarial Training

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1 Introduction

Deep learning (DL) has demonstrated remarkable potential to drive transformative advancements across a wide range of industries, such as autonomous driving and medical diagnostics. In such safety-critical domains, robustness is a fundamental prerequisite for DL models' pervasive deployment. Numerous studies have investigated DL robustness, leading to a range of benchmarks that systematically track the progress in the topic [8, 13, 22, 33, 34, 47]. These

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efforts have predominantly focused on *adversarial robustness* (AR), which emphasizes the extreme *worst-case* scenarios by evaluating local robustness based on the existence of *deterministic* adversarial examples (AEs)¹: subtle perturbations to inputs that alter model predictions [18, 41, 46].

A more recent and practical perspective, *probabilistic robustness* (PR) [4, 7, 14, 24, 42, 45, 48, 49, 54, 55, 62–67, 69], employs statistical approaches to answer the question: “What is the probability that predictions remain correct under stochastic perturbations”. This probabilistic view is arguably more relevant to real-world applications than AR, as it provides an *overall* assessment of a model’s local robustness, accounting for scenarios where AEs may exist [24, 54, 68] and acknowledging *residual risks* [64–66] that are more realistic to manage in practice.

Despite its potential, most existing work on PR is restricted to *evaluation*, with only a few studies [45, 52, 65, 66] exploring *training* methods specifically designed to *improve* PR [68]. Among them, we identify three limitations. First, they use different sets of evaluation metrics, with none adopting a comprehensive set to assess the training methods holistically, making it difficult to understand which methods are truly effective/efficient and under what conditions. Second, an interesting observation is that adversarial training (AT) that primarily designed to improve AR often enhances PR as a “by-product”. However, existing studies assess only a limited selection of AT methods, limiting the understanding of AT’s potential as an efficient PR improvement tool. Third, there is no *unified theoretical framework* to compare the generalisability of PR-targeted training, making it unclear of their broader applicability in various context. These gaps, together with the rapid growth of PR research highlight the need for a systematic benchmark for robust training methods targeting PR, which is currently missing from the literature.

To bridge the gaps, we introduce PRBench, the first benchmark dedicated to evaluating training methods for improving PR. Key features of PRBench include: *i)* A comprehensive set of metrics, covering clean accuracy, PR and AR performance, training efficiency, and generalisation error (GE); *ii)* A large set of training methods including most common AT methods and all PR-targeted training methods; *iii)* Theoretical bounds of GE are derived under the Uniform Algorithmic Stability Theory [6, 57] to conclude comparative insights. Specifically, PRBench includes 229 trained models based on 7 widely adopted datasets and 10 model architectures. It uses 4 common adversarial attacks to measure AR performance, 2 types of PR metrics (with various hyperparameters, e.g., distributions of perturbations) for PR, GE metrics for both AR and PR, the clean accuracy and the training time. While the total number of training methods included is 13, PRBench code-base is designed to be extendable for future inclusion and comparison of new methods.

Our analysis shows that, in most cases, AT surprisingly outperforms PR-targeted training in improving both AR and PR (not vice versa). That said, PR-targeted methods offer advantages such as better generalizability and higher clean accuracy, albeit often at

the expense of AR performance. The very recent “hybrid” training method combines the advantages of both, but at the price of training efficiency. At first glance, the observation that PR gains emerge when applying AT for AR may suggest that PR comes “for free” under AT. However, this is not meant literally, as trade-offs such as reduced clean accuracy and generalizability still apply. Without a dedicated PR benchmark, this “PR-for-free” hypothesis could not have been rigorously tested. Given the growing consensus that PR provides a more practical robustness perspective [52, 54, 64, 66, 68], PRBench resolves an open question regarding how PR should be incorporated into comprehensive evaluation: a research gap that could not have been addressed without PRBench. These findings further inspire future research directions: subsequent work may benefit from developing versatile, AT-based approaches that achieve balanced performance by jointly enhancing AR and PR while maintaining clean accuracy and low GE.

Key contributions of this paper include:

- **Benchmarking:** After formalizing a general formulation of the PR-targeted training methods, we develop the first benchmark dedicated for PR, evaluating a broad set of training methods with metrics of AR, PR, accuracy, generalisability, and efficiency.
- **Analysis:** Analysis is provided based on both empirically and theoretically, highlighting the strengths, limitations, and trade-offs among training methods in the context of improving PR.
- **Open-source repository and leaderboard:** All experimental details are at <https://github.com/wellzline/PRBenchmark>. A public leaderboard is available at <https://wellzline.github.io/PRBenchLeaderboard/>.

2 Preliminaries and Related Works

2.1 Generalisation Errors and Adversarial Robustness

Consider a classification task where $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^d$ denotes the inputs, and $y \in \mathcal{Y} \subseteq \{1, 2, \dots, \kappa\}$ represents the labels. Let $\mathcal{D}$ be an unknown probability measure over $\mathcal{X} \times \mathcal{Y}$. We define $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^{\kappa}$ as a DL model, parameterised by $\theta \in \Theta$, and $\mathbf{p}$ denote the *softmax* function. Given i.i.d. samples $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ drawn from $\mathcal{D}$ and a loss function² $\mathcal{L} : [0, 1]^{\kappa} \times \mathcal{Y} \rightarrow \mathbb{R}^+$, the *natural* and *empirical risks* can be represented as:

$$R(\theta) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [\mathcal{L}(\mathbf{p}(f(\mathbf{x}, \theta)), y)] \quad (1)$$

$$R_S(\theta) = \frac{1}{n} \sum_{i=1}^n \mathcal{L}(\mathbf{p}(f(\mathbf{x}_i, \theta)), y_i) \quad (2)$$

DEFINITION 1 (GENERALIZATION ERROR). *The GE of f on S is then defined as the difference between the natural and empirical risk:*

$$\text{GE}(\theta) = R(\theta) - R_S(\theta). \quad (3)$$

Although definitions of robustness vary across different DL tasks and model types, it generally refers to a DL model’s ability to maintain consistent predictions despite small input perturbations. Typically it is defined as all inputs in a region η have the same prediction, where η is a small norm ball (in a L_p -norm distance) of radius γ around an input $\mathbf{x}$. A perturbed input (e.g., by adding

¹The terms “worst-case” and “deterministic” indicate that AR studies typically aim to find the optimized AE, e.g., the one that maximizes loss or lies closest to the original input within a norm-ball. Note, although some approaches used to detect such deterministic/worst-case AEs involve *stochastic* procedures (e.g., random starts in PGD attacks), these are normally algorithmic choices for implementing numerical optimisation, which does not change the fact that the underlying optimal AE is deterministic.

²For simplicity, we also denote the composed loss function in Eq. 1 and 4 as $\mathcal{L}(\mathbf{x} + \delta, y; \theta)$ in the following sections, whenever it is unambiguous.

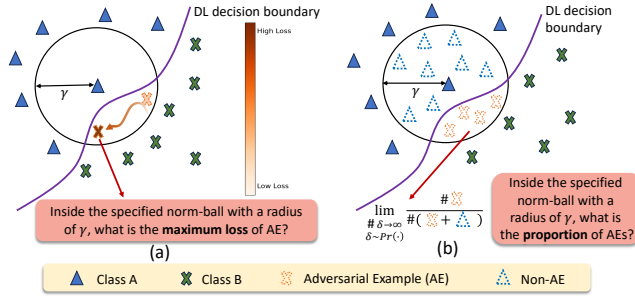


Figure 1: Comparison of Adversarial (a) and Probabilistic Robustness (b).

noise on x) x' within η is an AE if its prediction label differs from ground truth label y .

To evaluate AR, we normally formulate it as a question of maximizing the prediction loss, cf. Figure 1 (a), where *adversarial attacks* are introduced to find such a worst-case AE through perturbation:

$$\delta^* = \arg \max_{\|\delta\| \leq \gamma} \mathcal{L}(x + \delta, y; \theta). \quad (4)$$

AT [3, 18, 20, 35, 51] is the most common and effective empirical approach for enhancing AR. It is typically formulated as a min-max optimization problem, where the inner max aims at finding the worst-case AE in Eq. 4, e.g., by FGSM [18] and PGD [35]. Then the model is trained to minimise the loss over these AEs derived from a training dataset $\mathcal{D}$:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta\| \leq \gamma} \mathcal{L}(x + \delta, y; \theta) \right]. \quad (5)$$

2.2 Probabilistic Robustness

PR adopts a *probabilistic* view, evaluating the *overall* local robustness in the presence of AEs, cf. Figure 1 (b). This probabilistic notion of robustness is acknowledged to be more practical than the worst-case AR, as many applications only require the risk of AEs to stay below an acceptable threshold, rather than eliminating them entirely [54, 64, 67, 68].

DEFINITION 2 (PROBABILISTIC ROBUSTNESS). For a DL classifier f_{θ} that takes input x and returns a prediction label, the PR of an input x in a norm ball of radius γ is:

$$PR(x, \gamma) = \mathbb{E}_{\delta \sim Pr(\cdot | x)} [I_{\{f_{\theta}(x + \delta) = y\}}(x + \delta)], \quad (6)$$

where $I_S(x)$ is an indicator function that equals 1 when S is true and 0 otherwise; $Pr(\cdot)$ is the local distribution of inputs representing how perturbations δ are generated, which is precisely the “input model” used by [54, 55, 67].

Def. 2 suggests that PR is the probability that the model prediction remains unchanged from a random perturbation x' . A “frequentist” interpretation of this expected probability is—it is the *limiting relative frequency* of perturbations where the output label is preserved, in an infinite sequence of independently generated perturbations [67]. In other words, the “proportion” of non-AEs in the infinite set of perturbed inputs. To evaluate PR, one of the

earliest works [54] introduced a useful black-box statistical estimator, especially for cases where PR is very high. Later, more efficient white-box statistical estimators were proposed in [49]. Additionally, PR has been extended to applications such as explainable AI [24] and text-to-image models [67]. For an overview of PR estimators, we refer readers to [68].

In contrast to the relatively extensive studies on PR estimators, research on training methods for PR remains scarce. To the best of our knowledge, four studies [45, 52, 65, 66] explicitly motivated to develop training methods for improving PR. The first three can be categorized as Risk-based Training (RT) methods, which shift away from the worst-case perspective towards a training paradigm based on statistical risks induced by distributional perturbations, incorporating *perturbation risk functions* into the objective. We formally introduce its general formulation as:

DEFINITION 3 (RISK-BASED TRAINING (RT)). Reusing the notations above, RT is to

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{R}_{\gamma}(\mathcal{L}(x + \delta, y; \theta), \delta \sim Pr(\cdot | x))], \quad (7)$$

where $\mathcal{R}_{\gamma}$ is the perturbation risk function that defines some statistical quantities of the loss over a distribution of perturbations $Pr(\cdot | x)$.

The specific choice of $\mathcal{R}_{\gamma}$ in RT methods varies: [52] uses the identity function, while [45, 65] adopt functions calculating the Conditional Value-at-Risk (CVaR) and Entropic Value-at-Risk (EVaR), respectively. Essentially, rather than training on optimized AEs like AT for AR, RT methods train on statistical risks quantified by sampling stochastic perturbations from the perturbation distribution.

A very recent idea of adapting AT for PR was proposed in [66], which does not follow RT paradigm of Def. 3. Instead, their approach aligns with the traditional AT formulation in Eq. 5. Through multi-start PGD attacks and decision boundary exploration, AT-PR identifies an optimal AE whose neighborhood constitutes the largest all-AE region. Intuitively, training over such an optimal AE would reduce the overall “proportion” of AEs in the local norm ball and thus improve PR. We classify it as “hybrid” given it is targeting PR but adapting AT by solving a new min-max problem. Both RT methods and the “hybrid” method AT-PR constitute the current class of PR-targeted method. Further details for all four training methods are provided in Appendix B. For clarity, the main abbreviations used in this paper are summarized in Table 1.

Table 1: Summary of abbreviations used in the paper.

Acronyms	Meaning
AR	Adversarial Robustness
AT	Adversarial Training
ERM	Empirical Risk Minimization
GE	Generalization Error
PR	Probabilistic Robustness
RT	Risk-based Training

2.3 Related Benchmarks

Several robustness platforms have been developed to support AR evaluation by implementing popular attack methods, such as FoolBox [43], AdverTorch [12], Cleverhans [40], AdvBox [19], ART [39],

SecML [36], DeepRobust [31], etc. In addition, some AR benchmarks have also been established [8, 13, 22, 33, 34, 47]. Despite these efforts, existing work remains confined to AR evaluation, overlooking the more practical perspective of PR. Consequently, they fail to evaluate current robust training methods in terms of their improvements in PR or to reveal the inherent multi-objective trade-off relationship, thereby hindering the development of a holistic understanding of robustness progress. Compared to the aforementioned benchmarks, PRBench provides a more comprehensive framework in several key aspects: (1) it is the first benchmark to emphasize the PR, formalizing a general formulation of the PR-targeted training methods along with a summary of commonly used evaluation metrics and associated training methods; (2) it includes an extensive evaluation of widely used models and datasets, covering representative training methods designed for both AR and PR, enabling direct comparisons under PR metrics and revealing the performance of these methods in both adversarial and probabilistic perspectives; and (3) it presents in-depth analyses of the GE from empirical and theoretical standpoints, which has not been explored in prior benchmarks. These insights help improve DL model robustness in practical applications by providing the first standardized benchmark targeting PR, complementing the widely studied area of AR.

2.4 Generalization and Robust Overfitting

Several lines of research have studied robust overfitting and generalization error in DL models. We compare our work with four relevant studies that examine generalization from different perspectives and theoretical foundations.

Scope and goal. Jiang et al. [25] investigate generalization in standard deep learning models, focusing on empirical correlations between complexity measures, such as VC-dimension, norm-based, and PAC-Bayes measures, and generalization. Kim et al. [28] conduct a large-scale empirical study of robust generalization under AT, analyzing margin-, smoothness-, flatness-, and gradient-based measures and their correlations with the GE. Yin et al. [60] study adversarial robust generalization through Rademacher complexity. Dziugaite et al. [16] examine how evaluation methodologies can obscure the successes and failures of different generalization measures and propose a distributional robustness framework. In contrast, our study focuses on PR, evaluating improvements in PR achieved by various robustness training methods and providing theoretical analysis of the GE across different training objectives.

Theoretical foundation. Jiang et al. [25] base their analysis on theoretically motivated complexity measures from generalization theory, including VC-dimension, norm-based, and PAC-Bayes measures. Kim et al. [28] extend this framework to AT by examining margin-, smoothness-, and flatness-based measures. Yin et al. [60] employ Rademacher complexity to analyze adversarial robust generalization. Dziugaite et al. [16] complement Jiang et al. [25] by using a distributional robustness framework to evaluate generalization across diverse experimental settings. In our work, we leverage η -approximate β -smoothness [57] to analyze how optimization objectives affect GE, highlighting connections between optimization and generalization under different robustness regimes.

3 Description of PRBench

All existing robustness benchmarks focus exclusively on AR [8, 13, 22, 33, 34, 47], while our PRBench is the first benchmark for PR. In this section, we provide a detailed description of all the models, datasets, training methods and metrics used in PRBench. We provide details on hardware configuration, hyperparameter selection and computational setup in Appendix A. All experiments are conducted using a total of four NVIDIA H100 GPUs.

3.1 Models and Datasets

The models under evaluation span 3 types of architectures: (1) plain CNNs; (2) residual CNNs; and (3) transformer-based models. We train a diverse set of models, including VGG-19, SimpleCNN, ResNet-18, ResNet-34, WRN-28-10, ViT (Small/Base/Large), and DeiT (Tiny/Small). In total, 229 models are trained on 7 datasets: MNIST, SVHN, CIFAR-10, CIFAR-100, CINIC-10, TinyImageNet, and ImageNet-50. The details of the models and datasets are summarized in Table 2 and Table 3.

3.2 Training Methods

As shown in Table 4, PRBench categorizes all training methods into 4 groups: standard training (i.e., empirical risk minimization (ERM)), 6 AT methods, 4 RT methods (with corruption training using uniform, Gaussian, and Laplace noise) and the “hybrid” method AT-PR.

For AT, we consider the following representative methods. The PGD attack is widely regarded as the standard approach, which minimizes the adversarial cross-entropy loss $\mathcal{L}_{CE}$. Building on this, logit pairing methods [26], including adversarial logit pairing (ALP) and clean logit pairing (CLP), augment the objective with a regularization term coupling natural and AEs. TRADES further introduces a KL-based regularization, defining its objective as a combination of the natural loss and the KL divergence $\mathcal{L}_{KL}$ between predictions on clean and adversarial inputs. MART follows the same KL-based regularization but augments it with the adversarial cross-entropy loss $\mathcal{L}_{CE}$, while additionally emphasizing misclassified examples through larger penalty weights. Overall, existing AT methods can be characterized by different uses of $\mathcal{L}_{CE}$ and $\mathcal{L}_{KL}$, either individually or in combination, for both AE generation and training. We identify a missing configuration and introduce KL-PGD, a combination that generates AEs using $\mathcal{L}_{KL}$ (as in TRADES) and trains the model with $\mathcal{L}_{CE}$ (as in PGD). KL-PGD is designed as a diagnostic variant to disentangle the effects of loss functions from perturbation generation strategies, thereby providing a controlled comparison point for systematically evaluating the relative contributions of different optimization objectives.

By instantiating the general formulation of RT training (Def. 3), three existing works have been proposed, corresponding to essentially two distinct approaches depending on how the perturbation risk function $\mathcal{R}_\gamma$ is realized. A simple augmentation-based strategy is introduced by [52], referred to as *corruption training*, where each input $\mathbf{x}_i$ is perturbed by a sample drawn from a distribution (uniform by default) within an ℓ_∞ ball of radius γ . Robey et al. [45] proposed a CVaR-based risk objective that emphasizes the tail of the

Table 2: Summary of model architectures and training paradigms used in PRBench.

Category	Characteristics	Model	Size	Training Paradigm	
				Normal	Pre-trained
Plain CNN	Sequential conv layers	4-layer CNN	1.1M	✓	
		VGG-19	137M	✓	
Residual CNN	Residual connections	ResNet-18	11.7M	✓	
		ResNet-34	21.8M	✓	
		Wide-ResNet-28-10	36.5M	✓	
Transformer	Self-attention	ViT-Small	22M		✓
		ViT-Base	86M		✓
		ViT-Large	307M		✓
		DeiT-Tiny	5M		✓
		DeiT-Small	22M		✓

Table 3: Summary of datasets used in PRBench.

Dataset	Training Size	Test Size	Resolution	Number of Classes
MNIST	60,000	10,000	28×28	10
SVHN	73,257	26,032	32×32	10
CIFAR-10	50,000	10,000	32×32	10
CIFAR-100	50,000	10,000	32×32	100
CINIC-10	90,000	90,000	32×32	10
TinyImageNet	100,000	10,000	64×64	200
ImageNet-50	64,000	2,500	224×224	50

Table 4: Loss functions and AE generation strategies for different training methods.

Type	Method	Loss Function	AE Generation
Standard	ERM	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}, \theta), y)$	–
AT	PGD	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y)$	$\delta_t = \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}_{t-1} + \delta_{t-1}, \theta), y))$
	MART	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y) + \lambda(1 - \mathbf{p}(y \mathbf{x})) \cdot \mathcal{L}_{KL}(\mathbf{p}(\mathbf{x}, \theta) \parallel \mathbf{p}(\mathbf{x} + \delta, \theta))$	
	ALP	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y) + \lambda \cdot \ \mathbf{p}(\mathbf{x} + \delta, \theta) - \mathbf{p}(\mathbf{x}, \theta)\ _2^2$	
	CLP	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}, \theta), y) + \lambda \cdot \ \mathbf{p}(\mathbf{x} + \delta, \theta) - \mathbf{p}(\mathbf{x}, \theta)\ _2^2$	
	TRADES	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}, \theta), y) + \lambda \cdot \mathcal{L}_{KL}(\mathbf{p}(\mathbf{x}, \theta) \parallel \mathbf{p}(\mathbf{x} + \delta, \theta))$	
RT	KL-PGD	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y)$	$\delta_t = \alpha \cdot \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}_{KL}(\mathbf{p}(\mathbf{x}_{t-1}, \theta) \parallel \mathbf{p}(\mathbf{x}_{t-1} + \delta_{t-1}, \theta)))$
	Corruption	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y)$	$\delta \sim \text{Pr}(\cdot \mid \mathbf{x})$
Hybrid	CVaR	$\text{CVaR}_{1-\rho}(\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \delta, \theta), y))$	
	AT-PR	$\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}, \theta), y)$	$\delta = \arg \max_{\delta \in \{\delta^{PGD}\}} k \quad \text{s.t. } \text{Pr}(\mathbf{x} + \delta, k) = 0$

loss distribution, thereby prioritizing samples with larger loss values. A similar work by [65] replaces CVaR with EVaR, which leverages the entire distribution while following the same training strategy via sampling³. The “hybrid” method AT-PR targets PR while following the spirit of AT, as summarized in Table 4. It introduces a new min-max optimization objective (Eq. 25). Through multi-start PGD attacks and decision boundary exploration, AT-PR selects a local optimal AE in the norm-ball ($\delta \in \{\delta^{PGD}\}$) whose neighborhood constitutes the largest all-AE region k , i.e., $\text{Pr}(\mathbf{x} + \delta, k) = 0$.

3.3 Evaluation Metrics

In addition to clean accuracy, PRBench evaluates the models across three core aspects: AR, PR, and GE, cf. Table 5. AR performance is assessed by classification accuracy under adversarial attacks (Eq. 4). PR is quantified using two metrics from the literature: $\text{PR}_{\mathcal{D}}(\gamma)$ [54], the average probability that correctly classified inputs $\mathbf{x}_i$ remain correct under perturbations of radius γ (cf. Def. 2), and $\text{ProbAcc}(\rho)$ [45], the fraction of inputs $\mathbf{x}$ whose $\text{PR}(\mathbf{x}, \gamma)$ exceeds a given threshold $1 - \rho$. For the two PR metrics, $|\mathcal{D}|$ denotes the number of test samples that are correctly classified by the model, whereas for AR, $|\mathcal{D}|$ refers to the total number of test samples. GE captures

³Given the conceptual similarity, we categorize both EVaR and CVaR under the CVaR category in Table 4.

Table 5: Evaluation metrics of AR, PR, and GE.

Type	AR	PR	GE
Metric	$AR = \frac{1}{ \mathcal{D} } \sum_{\mathbf{x}_i \in \mathcal{D}} I_{\{f_{\theta}(\mathbf{x}'_i)=y_i\}}(\mathbf{x}_i)$	$PR_{\mathcal{D}}(\gamma) = \frac{1}{ \mathcal{D} } \sum_{\mathbf{x}_i \in \mathcal{D}} PR(\mathbf{x}_i, \gamma)$	$GE_{PR_{\mathcal{D}}(\gamma)} = PR_{\mathcal{D}_{\text{train}}}(\gamma) - PR_{\mathcal{D}_{\text{test}}}(\gamma)$
		$ProbAcc(\rho) = \frac{1}{ \mathcal{D} } \sum_{\mathbf{x}_i \in \mathcal{D}} I_{\{PR(\mathbf{x}_i, \gamma) \geq 1-\rho\}}(\mathbf{x}_i)$	$GE_{AR} = AR_{\mathcal{D}_{\text{train}}} - AR_{\mathcal{D}_{\text{test}}}$

robustness gap between the training and test sets. More details are deferred to Appendix C.

4 Theoretical Foundations of Robust Overfitting

While AT is regarded as one of the most promising methods for enhancing AR, empirical studies [21, 44] have shown that it suffers significantly from overfitting. To understand this phenomenon, several theoretical studies have been conducted under different frameworks, including VC-dimension [1, 37], Rademacher complexity [2, 27, 56, 60], and Uniform Algorithmic Stability [17, 23, 58, 59]. Moreover, Jiang et al. [25] study generalization in standard DL models, focusing on the empirical correlation between complexity measures such as VC-dimension and norm-based measures. Kim et al. [28] further conduct an empirical study of robust generalization under AT, analyzing how margin-based, smoothness-based, flatness-based, and gradient-based measures correlate with the generalization. Among all these theoretical perspectives, one of the most notable contributions is by [57], which introduces the concept of η -approximate β -smoothness, as defined in Eq. 8. This work [57] analyses the surrogate loss of the max function and show that generalization is affected by an additional term proportional to η .

DEFINITION 4 (APPROXIMATE SMOOTHNESS [57]). Let $f : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$, then f is η -approximate β -smooth, if for all $\mathbf{z} \in \mathbb{R}^d$, and $\forall \theta_1, \theta_2$, we have:

$$\|\nabla_{\theta} f(\mathbf{z}, \theta_2) - \nabla_{\theta} f(\mathbf{z}, \theta_1)\| \leq \beta \|\theta_2 - \theta_1\| + \eta. \quad (8)$$

To facilitate comparison between different AT schemes, we make the following assumptions.

ASSUMPTION 1. Assume that the machine learning model $f(\mathbf{x}, \theta)$ is L_{θ} -Lipschitz w.r.t. θ and L -Lipschitz w.r.t. $\mathbf{x}$ such that:

$$L_{\theta} \triangleq \sup_{\theta_2 \neq \theta_1} \frac{\|f(\mathbf{x}, \theta_2) - f(\mathbf{x}, \theta_1)\|}{\|\theta_2 - \theta_1\|} \quad (9)$$

$$L \triangleq \sup_{\mathbf{x}_2 \neq \mathbf{x}_1} \frac{\|f(\mathbf{x}_2, \theta) - f(\mathbf{x}_1, \theta)\|}{\|\mathbf{x}_2 - \mathbf{x}_1\|} \quad (10)$$

Similarly, we assume the smoothness condition for the gradient of model f w.r.t. θ as:

$$\forall \mathbf{x} \in \mathcal{X}, \quad \|\nabla_{\theta} f(\mathbf{x}, \theta_2) - \nabla_{\theta} f(\mathbf{x}, \theta_1)\| \leq \beta_{\theta} \|\theta_2 - \theta_1\|, \quad (11)$$

$$\forall \theta \in \Theta, \quad \|\nabla_{\theta} f(\mathbf{x}_2, \theta) - \nabla_{\theta} f(\mathbf{x}_1, \theta)\| \leq \beta \|\mathbf{x}_2 - \mathbf{x}_1\|. \quad (12)$$

This assumption has been widely adopted and validated in prior work [17, 57–59], where it serves as a foundational premise for analyzing algorithmic stability and generalisation under AT. We show that the GE is bounded in Theorem 1.

THEOREM 1. Given the Lipschitz and smoothness assumption in Assumption 1 for classifier f , we show that the surrogate loss

$$\max_{\|\delta\| \leq \gamma} \mathcal{L}_{CE}(\mathbf{p}(f(\mathbf{x} + \delta, \theta)), y) \quad (13)$$

is φ -Lipschitz and ϕ -approximate ψ -smooth, such that

$$\varphi = 2L_{\theta} \quad \text{and} \quad \phi = (4\beta\gamma + 2L_{\theta}) \quad \text{and} \quad \psi = (2\beta_{\theta} + L_{\theta}^2). \quad (14)$$

We run SGD with learning rate $\alpha_t \leq c/t$ for T steps with a constant c such that $1/c \geq \psi$. Then, the GE is bounded as

$$|R(\theta) - R_S(\theta)| \leq \frac{1}{n} + \frac{2\varphi^2 + n\phi}{\psi(n-1)} T. \quad (15)$$

Theorem 1 follows from the results in [57], where the authors conclude that the additional term $n\phi$ contributes to robust overfitting. The complete proof corresponds to the special case $\lambda = 0$ of Theorem 3 in Appendix D.2. And the proof for uniform stability is shown in Appendix D.5.

THEOREM 2. Follow the same condition for Theorem 1, consider the objective function for contained AT

$$\max_{\|\delta\|_2 \leq \gamma} \mathcal{L}_{CE}(\mathbf{p}(f(\mathbf{x} + \delta, \theta)), y) + \lambda \|\mathbf{p}(f(\mathbf{x} + \delta, \theta)) - \mathbf{p}(f(\mathbf{x}, \theta))\|_2^2. \quad (16)$$

We show that the Lipschitz constant, $\tilde{\varphi}$, for the objective function is

$$\tilde{\varphi} = 2L_{\theta} + 2\lambda(\nu\beta\gamma + 3\nu^2L_{\theta}) = \varphi + 2\lambda(\nu\beta\gamma + 3\nu^2L_{\theta}), \quad (17)$$

and it is $\tilde{\phi}$ -approximate $\tilde{\psi}$ -smooth, such that

$$\tilde{\phi} = (4\beta\gamma + 2\nu L_{\theta}) + 2\lambda\gamma(\nu^2\beta + 2\nu L_{\theta}) \quad (18)$$

$$= \phi + 2\lambda\gamma(\nu^2\beta + 2\nu L_{\theta}) - 2L_{\theta}(1 - \nu) \quad (19)$$

$$\tilde{\psi} = (2\beta_{\theta} + L_{\theta}^2) + 6\lambda(\nu^2\beta_{\theta} + 4\nu L_{\theta}^2) = \psi + 6\lambda(\nu^2\beta_{\theta} + 4\nu L_{\theta}^2) \quad (20)$$

where φ , ϕ and ψ are corresponding variables for AT in Theorem 1 without constraints and

$$\nu \triangleq \max_{\|\delta\|_2 \leq \gamma, \theta \in \Theta} \|\mathbf{p}(f(\mathbf{x} + \delta, \theta)) - \mathbf{p}(f(\mathbf{x}, \theta))\|_2, \quad (21)$$

denotes the upper bound of the penalty function.

Theorem 1 aims to establish the fundamental theoretical basis for our GE analysis: it formalizes the Lipschitzness and smoothness assumption of the adversarial surrogate loss follows from the results in [57], which is the key prerequisite for analyzing robust overfitting. Theorem 2 builds directly on Theorem 1 by extending these properties to the AT objective with regularization, quantifying how the added regularizer changes the Lipschitz and smoothness constants and thereby affects the GE behavior observed in our subsequent empirical results (see empirical analysis in Sec. 5.2).

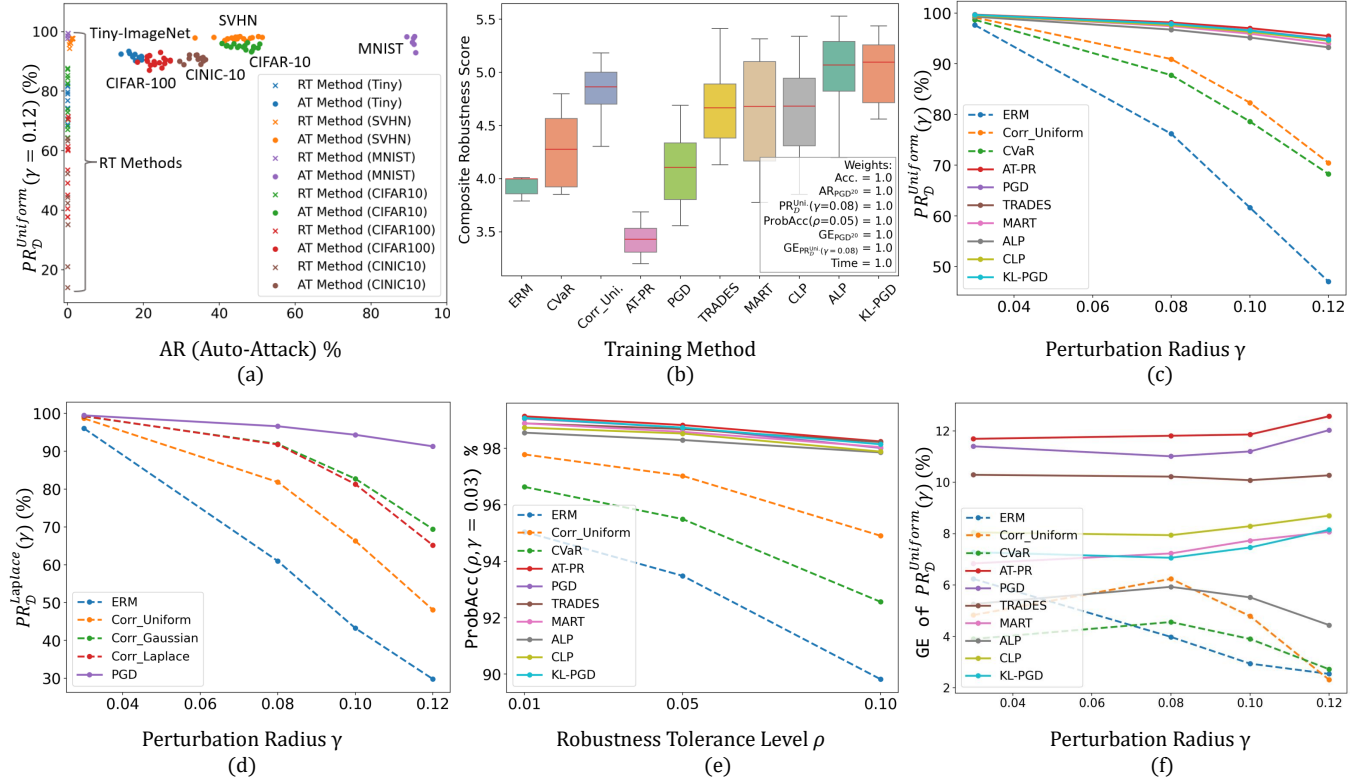


Figure 2: (a) Comparison of training methods (AT and RT) in terms of AR (AA) and PR ($PR_D^{\text{Uniform}}(\gamma)$) performance across various datasets. (b) Composite robustness scores of different training methods, aggregated over all test datasets and model architectures. (c) $PR_D^{\text{Uniform}}(\gamma)$ of ResNet-18 trained with different training methods on CIFAR-10 under varying γ . (d) $PR_D^{\text{Laplace}}(\gamma)$ for ResNet-18 trained with corruption training and PGD models on CIFAR-10 across various γ . (e) $ProbAcc(\rho, \gamma = 0.03)$ for ResNet-18 trained with different training methods on CIFAR-10 with respect to different robustness tolerance level ρ . (f) GE of $PR_D^{\text{Uniform}}(\gamma)$ for ResNet-18 trained with different training methods on CIFAR-10 with respect to different γ . More experimental results are deferred to Appendix E.

5 Analysis and Discussion

5.1 Empirical Analysis of Evaluation Results

While the complete evaluation results across all models and datasets are provided in Appendix E, we present *representative* example in Tables 6 and 7 and Figure 2 for brevity.

1. AT methods effectively improve both AR and PR, at the cost of reduced clean accuracy and generalization. All AT methods consistently improve PR alongside AR. Within the training perturbation norm-ball, models trained by AT achieve over 99% of $PR_D(\gamma)$, cf. Table 6. The $ProbAcc(\rho = 0.01)$ metric further indicates that more than 97% of test samples exhibit $PR_D(\gamma) > 99\%$. Moreover, as shown in Figure 2 (c) (e), model trained with AT methods remains stable PR performance as the perturbation radius γ increases and across different robustness tolerance levels ρ , consistently achieving over 93% robustness under both $PR_D(\gamma)$ and $ProbAcc(\rho)$. Our results show that AT, although originally designed to improve AR, consistently yields improved PR as a “by-product”. However, it does not imply there is no cost in other metrics, indeed, AT still suffers from a drop in clean accuracy. For example, the clean accuracy decreases from 94.85% under standard training to 83.83%

after PGD training (cf. Table 6). In addition, AT methods are prone to robust overfitting, as reflected by large GE. For instance, the model trained with PGD exhibits an AR GE of 25.02%. In this sense, the phenomenon can be interpreted as a multi-objective trade-off.

2. RT methods underperform AT methods on PR performance. While RT methods effectively improve PR compared with ERM, they still lag behind AT methods, as shown in Table 6. Moreover, their PR performance drops sharply as the perturbation radius increases, cf. Figure 2 (c) (d), showing limited generalization beyond the norm-ball setting in training. Notably, our analysis of the CVaR codebase from [45] suggests a potential evaluation inconsistency: it includes misclassified original clean samples when computing $ProbAcc(\rho)$ rather than focusing on the robustness of correctly classified original samples. We have corrected this in our experiments. This view is also supported by prior studies, e.g., [32], which states: “When considering the adversarial robustness of a wrongly classified sample x , the robustness should be 0”. Similarly, [5] emphasizes that “AEs must satisfy the constraints: the original examples are classified correctly while the predictions of the AEs are wrong”. In other words, if a sample is misclassified by the

model initially, its ability to resist perturbations and maintain the wrong prediction (i.e., being “robustly wrong”) does not constitute meaningful robustness.

3. AT methods ensure stable and high PR under diverse perturbation distributions. We evaluate the PR performance of PGD-trained models under three common noise distributions: Uniform, Gaussian, and Laplace, and compare them with models trained by Corruption method using the same noise distribution. As shown in Table 7 and Figure 2 (d), PGD-based AT consistently achieves the highest PR across all distributions, outperforming Corruption in all settings.

4. RT methods achieve higher clean accuracy but has near-zero improvement on AR As shown in Table 6, models trained with RT methods exhibit smaller reductions in clean accuracy. However, their AR degrades significantly, with performance dropping to nearly zero under strong attacks such as C&W and Auto-Attack. Only the CVaR method achieves non-zero AR performance against PGD attacks, though it still lags behind models trained with AT methods. A similar observation is evident in Figure 2 (a), where RT methods consistently appear along the leftmost vertical axis, indicating near-zero AR performance across all datasets and models. More results are shown in Table 10.

5. AT-PR achieves a better trade-off among Accuracy, AR, and PR. As shown in Table 6: *i*) it achieves comparable PR performance to AT methods; *ii*) it achieves AR slightly weaker than AT, yet still substantially higher than that of RT methods; *iii*) it reduces clean accuracy less severely than AT. Overall, although AT-PR is designed as a PR-targeted method, its AT-inspired formulation (min-max optimization) delivers a more balanced performance across the three metrics.

6. Training efficiency varies widely. Table 6 shows that Corruption method is highly efficient, generating only a single random sample per training example, whereas CVaR is computationally expensive due to the need for multiple samples to approximate the perturbation distribution. Standard AT methods exhibit intermediate and more consistent training times. AT-PR is the most costly, as it generates a set of PGD-based AEs and selects the one maximizing the all-AE region, leading to a substantial increase in training time (as noted by the complexity analysis in [66]). To further investigate efficiency-oriented approaches, we additionally integrate the filtering-based method of [5] into PRBench as a case study (Table 13). This aligns with ongoing work on efficient robustness, and our extensible design enables inclusion of future methods.

7. AT methods lead in composite robustness evaluation. To holistically assess training methods across all key aspects (AR, PR, GE, clean accuracy, and training efficiency), we introduce a *composite robustness score*. This score is computed as a (weighted) sum of Min–Max normalized metric values, with “lower-is-better” metrics (e.g., GE, training time) reversed for consistency. Figure 2 (b) presents results under equal-weighting scheme, where AT methods consistently rank highest, corroborating earlier findings that they generally outperform PR-targeted training methods. Moreover, the KL-PGD achieves competitive performance, indicating that generating AEs with $\mathcal{L}_{KL}$ is more effective than with the traditional $\mathcal{L}_{CE}$. For additional results see Figure 6.

5.2 Generalization Error Analysis

The empirical results reveal several insights into the GE of different training methods. In this section, we provide a theoretical analysis to support the empirical findings.

1. RT methods consistently yield lower GE. As shown in Table 6 and Figure 2 (f), all RT methods empirically exhibit smaller GEs than AT methods across both AR and PR metrics. We next investigate the theoretical basis of this generalizability advantage in the Proposition 1.

PROPOSITION 1. *Let $\mathcal{L}$ be the objective loss function for training without adversarial perturbation, and assume that $\mathcal{L}$ is ϕ -Lipschitz and ψ -smooth. Then, the objective function of the CVaR-based training scheme in Eq. 7 is also ϕ -Lipschitz and $\max\{\phi, \psi\}$ -smooth.*

Our Proposition 1 shows that RT learning yields a smoother training process than AT, as it excludes the GE term (ϕ in Theorem 1) associated with robust overfitting, resulting in a smaller overall GE. Training with corruption can be seen as a special case of CVaR and thus inherits the same Lipschitz and smooth properties. The proof of our Proposition 1 is in Appendix D.3.

As shown in Theorem 2, successful training with a properly chosen λ can effectively reduce the L_2 distance between softmax outputs, denoted by v . This reduction pushes $\tilde{\varphi} \approx \varphi$ and $\tilde{\psi} \approx \psi$, and in cases where v becomes sufficiently small, the training process may become even smoother, potentially satisfying $\tilde{\phi} < \phi$. This provides a theoretical explanation for why AT methods that impose constraints on the softmax outputs $\mathbf{p}$ tend to exhibit smaller GEs. On the other hand, an improperly chosen λ can result in highly non-smooth optimization. The detailed proof for our Theorem 2 is in Appendix D.2. We now provide theoretical insights into the following two empirical findings.

2. PGD has the highest GE among AT methods: Within AT methods (Table. 6), PGD results in the largest GE under the GE_{AR} metric. This is due to the absence of an explicit penalty in its objective, unlike methods such as TRADES and MART, which incorporate additional penalties to stabilize training and improve generalization. This empirical result corresponds to our theoretical analysis in Theorem 2. The AT with a penalty over the distance between softmax before and after perturbation effectively enhance the smoothness for the objective function. Additional results in Appendix E.

3. AT-PR exhibits similar GE to PGD. AT-PR, though designed as a PR-targeted method, is AT-inspired: it selects AEs from a set of PGD-generated candidates corresponding to different local optima. Consequently, as shown in Table 6 & 10, its GE aligns closely with that of PGD. A detailed theoretical analysis with its corresponding algorithm is provided in Appendix D.4.

4. Optimization budget explains the GE gap between ALP and CLP: CLP shows higher GE than ALP, as ALP uses AEs in its optimization objective while CLP relies on clean inputs. This suggests that ALP’s higher optimization budget leads to better generalization. According to Theorem 2, the generalization disadvantage of CLP arises from the choice of λ . To ensure robustness against adversarial attacks, CLP requires a relatively large value of λ , since it is based on $\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x}, \boldsymbol{\theta}), y)$ instead of $\mathcal{L}_{CE}(\mathbf{p}(\mathbf{x} + \boldsymbol{\delta}, \boldsymbol{\theta}), y)$, cf. Table 4. This leads to reduced smoothness and a larger GE.

Table 6: Performance of different training methods on CIFAR-10 (ResNet-18), evaluated by clean accuracy (Acc.), AR, PR, GE performance, and training time (sec./epoch).

Type	Method	Acc. %	AR %				$\text{PR}_{\mathcal{D}}^{\text{Uniform}}(\gamma) \%$				$\text{ProbAcc}(\rho, \gamma = 0.03) \%$			$\text{GE}_{\text{AR}} \%$	$\text{GE}_{\text{PR}_{\mathcal{D}}^{\text{Uni}}(\gamma)} \%$				Time
			PGD^{10}	PGD^{20}	CW^{20}	AA	0.03	0.08	0.1	0.12	0.1	0.05	0.01	PGD^{20}	0.03	0.08	0.1	0.12	
Std.	ERM	94.85	0.01	0.0	0.0	0.0	97.64	76.19	61.65	47.07	95.04	93.48	89.82	0.0	6.24	3.98	2.94	2.54	3
RT	Corr_Uniform	94.17	0.28	0.05	0.02	0.0	99.12	90.92	82.32	70.48	97.78	97.02	94.9	0.04	4.83	6.24	4.79	2.31	3
	CVaR	89.91	41.79	33.45	0.0	0.0	98.67	87.75	78.62	68.27	96.63	95.49	92.56	1.13	3.9	4.56	3.9	2.72	61
Hybrid	AT-PR	86.35	48.22	46.53	47.39	44.01	99.68	98.13	97.01	95.46	99.13	98.82	98.24	27.43	11.69	11.81	11.86	12.57	160
AT	PGD	83.83	50.86	49.46	49.18	46.34	99.63	97.89	96.59	94.85	99.05	98.73	98.01	25.02	11.4	11.01	11.2	12.03	30
	TRADES	83.34	54.09	53.15	50.84	48.99	99.55	97.74	96.33	94.55	98.88	98.68	98.22	16.12	10.29	10.22	10.08	10.27	25
	MART	82.26	54.83	53.84	49.63	46.94	99.56	97.4	95.92	93.93	98.88	98.58	98.04	17.4	6.84	7.23	7.73	8.07	22
	ALP	73.98	54.41	54.08	50.72	48.41	99.27	96.73	95.16	93.24	98.55	98.29	97.85	8.44	5.26	5.93	5.52	4.44	36
	CLP	81.47	54.12	53.34	51.05	49.05	99.53	97.61	96.29	94.54	98.73	98.52	97.88	15.32	8.05	7.94	8.29	8.7	36
	KL-PGD	87.55	49.4	48.43	47.06	44.77	99.63	97.86	96.54	94.72	99.07	98.73	98.14	14.57	7.28	7.06	7.46	8.15	27

Table 7: Comparison of ERM, PGD, and Corruption-trained models under Uniform, Gaussian, and Laplace noise. Reports accuracy (Acc.), PR, and GE at various perturbation radii γ on CIFAR-10 (ResNet-18).

Type	Method	Acc. %	$\text{PR}_{\mathcal{D}}^{\text{Uniform}}(\gamma) \%$				$\text{PR}_{\mathcal{D}}^{\text{Gaussian}}(\gamma) \%$				$\text{PR}_{\mathcal{D}}^{\text{Laplace}}(\gamma) \%$				$\text{GE}_{\text{PR}_{\mathcal{D}}(\gamma=0.03)} \%$		
			0.03	0.08	0.1	0.12	0.03	0.08	0.1	0.12	0.03	0.08	0.1	0.12	Uni.	Gau.	Lap.
Std	ERM	94.85	97.64	76.19	61.65	47.07	96.16	61.88	44.2	30.6	96.04	60.98	43.24	29.79	6.24	6.36	6.3
RT	Corr_Uniform	94.17	99.12	90.92	82.32	70.48	98.69	82.46	67.4	49.32	98.67	81.85	66.29	48.09	4.83	5.03	5.05
	Corr_Gaussian	93.32	99.41	96.22	92.14	85.32	99.23	92.23	83.42	70.46	99.22	91.91	82.72	69.42	5.69	5.67	5.6
	Corr_Laplace	93.61	99.45	96.32	91.97	84.4	99.3	92.09	82.09	66.42	99.29	91.78	81.3	65.16	5.72	5.67	5.63
AT	PGD	83.83	99.63	97.89	96.59	94.85	99.48	96.68	94.5	91.54	99.47	96.59	94.34	91.29	11.4	11.24	11.21

6 Conclusion

We present PRBench, the first dedicated benchmark for PR, designed to standardize the comprehensive evaluation of robustness training methods. It aims to advance our understanding of PR improvement from a broader perspective, beyond optimizing PR or AR in isolation, while accounting for the inherent trade-offs with clean accuracy and generalization.

Through a systematic evaluation of 229 models across diverse architectures, datasets, training methods, two distinct trade-off frontiers are identified. The first, represented by AT, emphasizes high AR and PR at the cost of clean accuracy and GE. The second, represented by RT, prioritizes high clean accuracy, PR and lower GE (supported by theoretical analysis) while trading off AR performance. In addition, we conduct an in-depth theoretical analysis of GE across all evaluated training methods to explore their generalization behavior, grounded in Uniform Algorithmic Stability.

At first glance, PRBench reveals that PR gains can emerge when applying AT for AR, which may give the *misleading impression* that “PR comes for free”. However, such gains are not without cost, as they often come at the expense of other metrics. Consequently, a dangerous misinterpretation of PRBench results is to conclude that PR-targeted methods are obsolete. Quite the opposite, we argue that future research on robustness training would benefit from more *versatile* approaches that seek balanced performance by jointly enhancing both AR and PR while maintaining clean accuracy and low GE. As such new generations of versatile robustness training methods emerge, PRBench will be essential for enabling fair comparison and evidence-based progress in robustness research.

In summary, as the robustness research community increasingly recognizes the importance of PR and the need to jointly improve both AR and PR in robustness training, PRBench addresses the open question of how PR should be incorporated into comprehensive robustness evaluation, a gap that could not be systematically studied without such a benchmark, thereby complementing the widely studied area of AR. As a benchmark, PRBench is intended to spark discussion and motivate rigorous future work to validate, refine, or refute emerging robustness training methods, by providing a foundation for systematic and evidence-driven analysis.

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A Hyperparameter Selection and Implementation Details

A.1 Experiment Setup

For the MNIST dataset, we employ a four-layer CNN, consisting of two convolutional layers followed by two fully connected layers, as described in [45]. For the CIFAR-10, CIFAR-100, CINIC-10 and SVHN datasets, we independently train VGG-19, ResNet-18, and WideResNet-28-10 on each dataset, and additionally include Vision Transformer (ViT) and Data-efficient Image Transformer (DeiT) for CIFAR-10. For the TinyImageNet dataset, we train ResNet-18 and ResNet-34 using the same training configuration as for CIFAR-10. For the ImageNet-50 dataset, we train a ResNet-18 model separately.

All models are trained using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay coefficient of 3.5×10^{-3} . Training is performed for 100 epochs (200 for ImageNet-50) with an initial learning rate of 0.01, which is decayed by a factor of 10 at epochs 75 and 90, following the setup in [45, 53].

A.2 Training Algorithms Hyperparameter Setting

- **PGD.** We set the perturbation radius to $\gamma = 8/255$ for all dataset except MNIST. During training, we performed 10 steps of projected gradient descent attack, using a step size of $\alpha = 2/255$ for CIFAR-10, CIFAR-100, CINIC-10, TinyImageNet and ImageNet-50, and a step size of $\alpha = 1.25/255$ for SVHN. For the MNIST dataset, the perturbation radius is set to $\gamma = 0.3$ with an attack step size of $\alpha = 0.1$, as MNIST is a relatively easier task that requires a larger budget, particularly for random perturbations.
- **TRADES.** We used the same step size and number of steps as described above for PGD. Additionally, we applied a weight of $\lambda = 6.0$ for all datasets, following the approach in [45, 61].
- **MART.** We used the same step size and number of steps as described above for PGD. Additionally, we applied a weight of $\lambda = 5.0$ for all datasets, following the approach in [53].
- **ALP.** Follow the work [26], we set $\lambda = 1$ for all datasets, except $\lambda = 0.01$ for SVHN, MNIST, ImageNet-50 and TinyImageNet.
- **CLP.** We set $\lambda = 1$ for all datasets, except $\lambda = 0.3$ for SVHN, MNIST, ImageNet-50 and TinyImageNet.
- **CVaR.** We set the number of perturbed samples drawn from the uniform distribution to 20, following the original setup in [45].
- **Corruption.** We perform standard corruption training by sampling a perturbation from a uniform distribution for each training point. We conduct corruption training using two other perturbation distributions: Gaussian and Laplace for comparison, with all perturbations constrained within the specified norm-ball radius.
- **AT-PR.** The number of PGD-based AE candidates is set to 5 while other parameters follow the original setup in [66].
- Following the standard efficient parameter config. [5], we integrate it into TRADES with $m = 1-8$ and $k = 10$ (10-step attack).

A.3 Model and Dataset Selection

In PRBench, we consider three major model architectures: (1) **plain CNNs**, such as VGG, representing early feedforward convolutional networks; (2) **residual CNNs**, exemplified by ResNet, which introduce skip connections to enable deeper architectures; and (3) **transformer-based models**, including Vision Transformer (ViT) [15] and Data-efficient Image Transformer (DeiT) [50], which leverage self-attention mechanisms for image representation learning. In our experiments, we conduct different training methods (See Table. 4) across a diverse set of models: VGG-19, SimpleCNN (4-layer CNN), ResNet-18, ResNet-34, Wide-ResNet-28-10, ViT, and DeiT, covering various scales such as ViT-Small, ViT-Base, ViT-Large, DeiT-Tiny, and DeiT-Small. See Table 2 for details.

Table 3 presents the seven widely used benchmark datasets used by PRBench, each representing a different level of complexity and image characteristics.

- **MNIST [11]** is a classic dataset of handwritten digits, consisting of 60,000 training images and 10,000 test images. Each image is a

28x28 pixel grayscale image, and the dataset is categorized into 10 classes (0-9).

- **SVHN [38]** (Street View House Numbers) is a real-world dataset derived from Google Street View house numbers. It contains 73,257 training and 26,032 test images, each sized 32x32 and labeled into 10 digit classes. Compared to MNIST, SVHN is more challenging due to background clutter and variation, making it suitable for evaluating robustness under real-world conditions.
- **CIFAR-10 [29]** is a widely used image classification dataset with 60,000 32x32 color images across 10 classes. The dataset is split into 50,000 training images and 10,000 test images. Compared to MNIST and SVHN, CIFAR-10 presents greater variability in content and background, making it a standard benchmark for evaluating generalization across diverse object classes.
- **CIFAR-100 [29]** extends CIFAR-10 by increasing the number of classes to 100, with each class containing 600 images. The dataset is also composed of 60,000 images (50,000 for training and 10,000 for testing) with 32x32 size. Its finer-grained classification task poses a greater challenge, requiring models to generalize across a broader and more diverse set of object categories.
- **CINIC-10 [9]** is derived from CIFAR-10 and downsampled ImageNet, with all images resized to 32x32 resolution. It contains 270,000 images evenly split into training, validation, and test sets (90,000 each). Like CIFAR-10, it has 10 classes, but the larger scale and more diverse image sources make it a more challenging and realistic benchmark.
- **TinyImageNet [30]** is a subset of the ImageNet dataset, containing 100,000 training, 10,000 validation, and 10,000 test images, all resized to 64x64 pixels and spanning 200 object classes. With higher resolution and a larger number of categories, TinyImageNet poses a more complex classification challenge, making it well-suited for evaluating model scalability and robustness in realistic settings.
- **ImageNet-50 [10]** is a 50-class subset of ImageNet. It consists of approximately 65,000 training images and 2,500 validation images at 224x224 resolution. Despite being smaller than full ImageNet, ImageNet-50 retains considerable diversity and complexity, serving as a scalable benchmark for robustness evaluation.

B PR-targeted training method

Some recent works have focused on designing training techniques specifically to improve PR [52] proposed a simple augmentation-based approach referred to as *corruption training*, where the model is trained on randomly perturbed examples. This process is equivalent to standard AT training, except that each input x_i is perturbed by a sample drawn from a uniform distribution within an ℓ_∞ norm ball of radius γ , rather than using worst-case AEs. They show that corruption training improves PR and reduces generalization error under PR evaluation settings. The corresponding optimization objective is formalized as:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(x + \delta, y; \theta)]; \delta \sim Pr(\cdot | x). \quad (22)$$

Later, [45] introduced an optimization objective based on the *Conditional Value-at-Risk (CVaR)* to improve PR. Specifically, for a

loss function $\mathcal{L}$ and a continuous distribution Pr , CVaR can be interpreted as the expected value of $\mathcal{L}$ over the tail of the distribution. It admits the following convex, variational characterization, which allows it to be optimized via stochastic gradient-based methods. Based on this, they formalized the training objective as:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\text{CVaR}_{1-\rho}(\mathcal{L}(x + \delta, y; \theta); \delta \sim Pr(\cdot | x))];$$

$$\text{CVaR}_{\rho}(\mathcal{L}; Pr) = \inf_{\alpha \in \mathbb{R}} \left[\alpha + \frac{\mathbb{E}_{\delta \sim Pr(\cdot | x)} [(\mathcal{L}(x + \delta) - \alpha)_+]}{1 - \rho} \right], \quad (23)$$

where $Pr(\cdot | x)$ is a uniform perturbation distribution, and $\rho \in (0, 1)$ determines the focus on the tail of the loss distribution. The threshold α is dynamically optimized during training to ignore perturbations with loss values below it, thereby encouraging the model to focus on AEs that incur higher but not necessarily maximal loss, aligning with the CVaR principle of prioritizing the worst-performing fraction of the distribution.

Building upon CVaR [65] propose Probabilistic Risk-averse Robust Learning with Stochastic Search (PRASS), which trains a risk-averse model by mitigating the *Entropic Value-at-Risk (EVAR)* over the captured worst-case perturbation distribution. Unlike CVaR, which focuses solely on the tail samples, EVAR further capitalizes on the entire distribution of samples. Specifically, EVAR with risk level $\rho \in (0, 1]$, denoted as $\text{EVAR}_{\rho}(\mathcal{L}(\delta))$, is defined as the infimum over $\alpha > 0$ of the Chernoff bound for $\mathcal{L}(\delta)$ with respect to the random variable δ . They then proposed a risk-averse robust learning paradigm to optimize the upper bound, and formalized the training objective as:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\text{EVAR}_{1-\rho}(\mathcal{L}(x + \delta, y; \theta); \delta \sim Pr(\cdot | x))];$$

$$\text{EVAR}_{\rho}(\mathcal{L}; Pr) = \inf_{\alpha > 0} \left[\frac{1}{\alpha} \log \left(\frac{\mathbb{E}_{\delta \sim Pr(\cdot | x)} [e^{\alpha \mathcal{L}(x + \delta)}]}{\rho} \right) \right], \quad (24)$$

thus the inner minimization over α is used to compute EVAR, while the outer minimization updates the model parameters.

The latest PR-targeted work [66], though aimed at PR, is not based on the perturbation risk function in Def. 3. Instead, it is closer to the traditional AT formulation in Eq. 5, introducing a new min-max objective designed to identify AEs that lie in the largest all-AE region k (i.e., $PR(x + \delta, k) = 0$). They achieve this by designing a numerical algorithm that first uses PGD to generate a diverse set of AEs at different local optima, i.e., $\mathcal{S} = \{x'_1, x'_2, \dots, x'_m\}$. For each x'_i , gradient descent is then performed toward the nearest decision boundary, and the AE with the largest distance to its nearest boundary is selected, corresponding to the largest all-AE region. Formally, for a model f_{θ} trained on dataset $\mathcal{D}$, the objective of AT-PR is:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta\| \leq \gamma, PR(x + \delta, k) = 0} k \right], \quad (25)$$

where k and δ are variables in the inner maximization, and $PR(\cdot, \cdot)$ is defined in Eq. 6. The maximization seeks an optimal δ^* producing an AE $x' = x + \delta^*$, and maximizes the radius k of a smaller norm-ball around x' where all inputs are AEs (i.e., $PR(\cdot, \cdot)$ within the region is 0).